

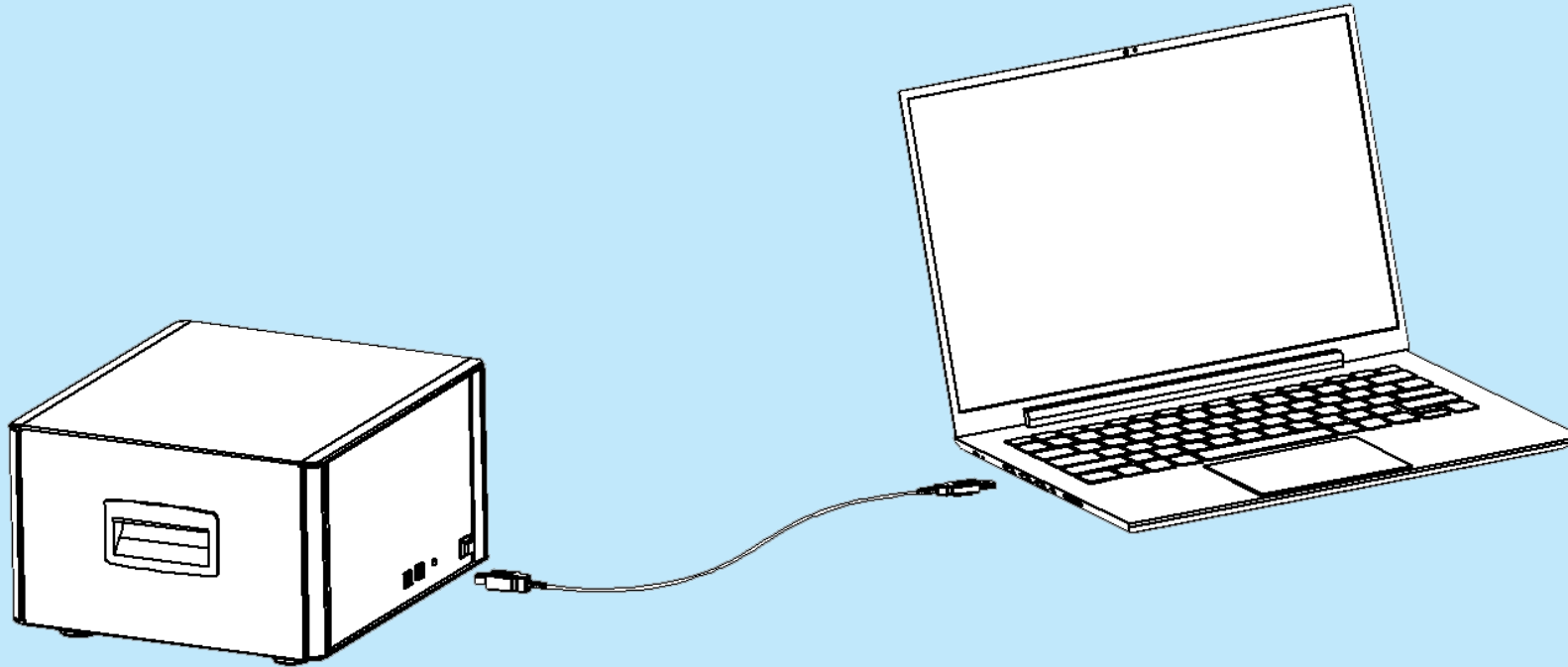


HD-SC™ Quick Start Guide

Version 2.2.1
Dec. 2019

STEP 1

- Connect device to included laptop



STEP 2

- Open software and log in as administrator
 - Default Username: admin
 - Password: admin123
- For operator use, see page 7
- Select connect

STEP 3

The screenshot displays the Soterixmedical software interface, which is divided into several functional areas:

- Settings Panel (Left):** Contains configuration options for stimulation. The 'Arbitrary' tab is active, showing settings for Individual Intensity (1.00 mA), Waveform (tACS), Duration (10 s), Delay (0 ms), Waveform Polarity (Bipola), SHAM (Off), Ramp up (0 s), Ramp down (0 s), and Ramp occurrence (Beginning). Below these are 'Channel Selection' options for Single or Multiple channels. A 'NET INTENSITY: 2.00 mA' label is visible at the bottom of this panel.
- Sample Preview (Center):** A graph showing the amplitude of the stimulation waveform over time (0 to 2000 samples). Two waveforms are plotted: Ch 1 (green) and Ch 6 (orange). The x-axis is labeled 'Sample' and the y-axis is labeled 'Amplitude'.
- Control Panel (Right):** Features a 'START' button for beginning stimulation, a 'LOAD' button for loading settings, and a 'Stimulation Status' section showing a '-' sign. Below this is a 'Preview Waveform' section.
- Selected Channels (Bottom Center):** A section showing the configuration for the selected channels. It lists Ch 1 and Ch 6, both set to 1.00 mA. Ch 1 has a frequency of 250 Hz and Ch 6 has a frequency of 1000 Hz. Below this, it shows '3K' and '2K' resistance measurements.

Annotations on the image point to specific features:

- 'Options to configure stimulation settings' points to the Settings Panel.
- 'Total Current' points to the 'NET INTENSITY: 2.00 mA' label.
- 'Intensities for selected channels' points to the 'Selected Channels' section.
- 'Shows all channels and resistance measurements' points to the 'Selected Channels' section.
- 'Begin Stimulation' points to the 'START' button.
- 'Load Settings' points to the 'LOAD' button.
- 'Preview Waveform' points to the 'Preview Waveform' section.

WAVEFORM OPTIONS: BASIC AND ARBITRARY

The 'Basic' tab is selected. The interface includes the following controls:

- Individual Intensity (mA): 1.00
- Waveform: tACS
- Duration (s): 10
- Delay (ms): 0
- Waveform Polarity: Bipola
- SHAM: Off
- Ramp up (s): 0
- Ramp down (s): 0
- Ramp occurrence: Beginning
- Channel Selection: A slider is positioned towards 'Single'.
- Single / Multiple: Radio buttons at the bottom.

Select waveform and intensity

Set stimulation duration

Set Waveform Polarity

Select duration

Select to enable sham

Set ramp conditions

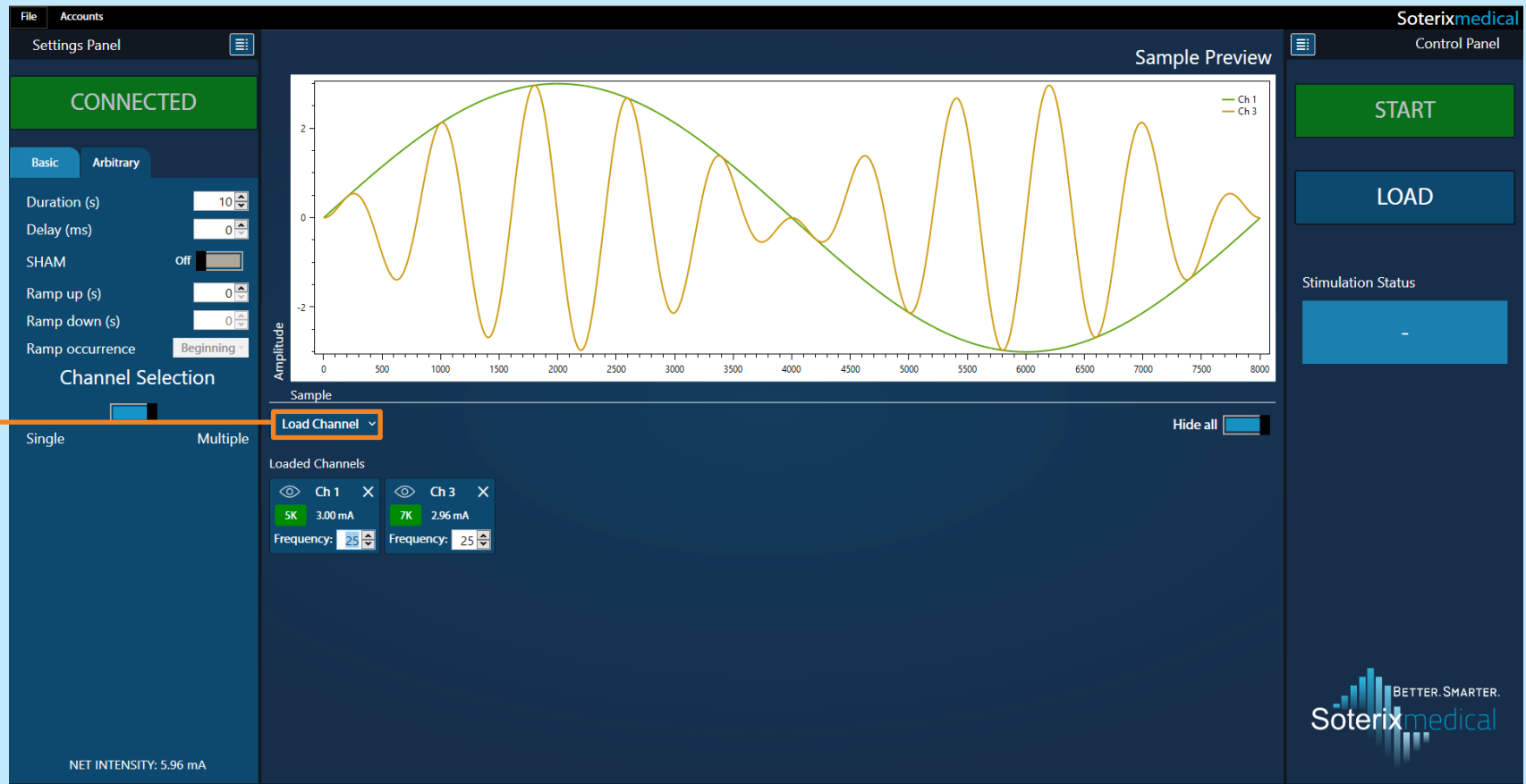
Selecting single or multiple waveforms across channels

The 'Arbitrary' tab is selected. The interface includes the following controls:

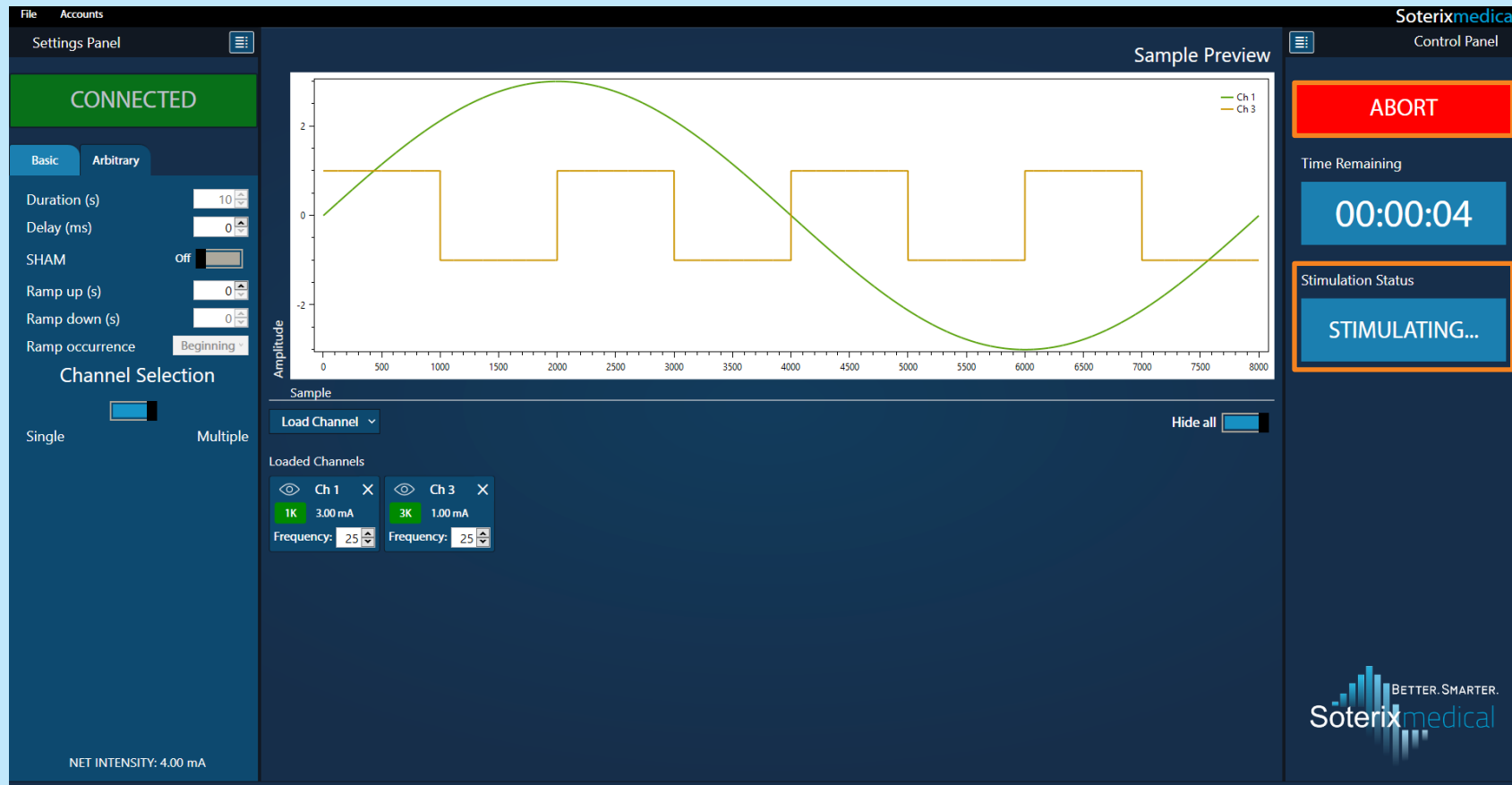
- Duration (s): 10
- Delay (ms): 0
- SHAM: Off
- Ramp up (s): 0
- Ramp down (s): 0
- Ramp occurrence: Beginning
- Channel Selection: A slider is positioned towards 'Single'.
- Single / Multiple: Radio buttons at the bottom.

ARBITRARY WAVEFORM

Click to select
arbitrary
waveform
from text file



STIMULATION WINDOW

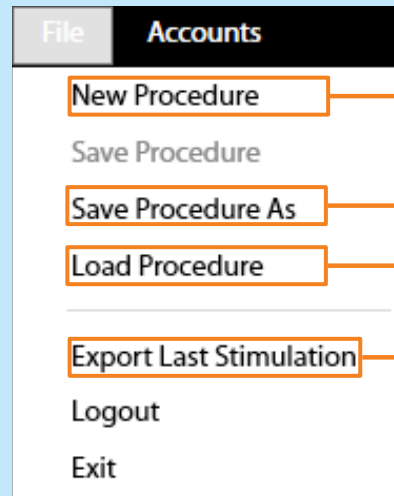


Manual
abort

Device
status

DOUBLE BLIND: ADMIN

- Default Username: admin
- Password: admin123



The 'Accounts' menu is open, showing several options. The following options are highlighted with orange boxes:

- New Procedure
- Save Procedure As
- Load Procedure
- Export Last Stimulation

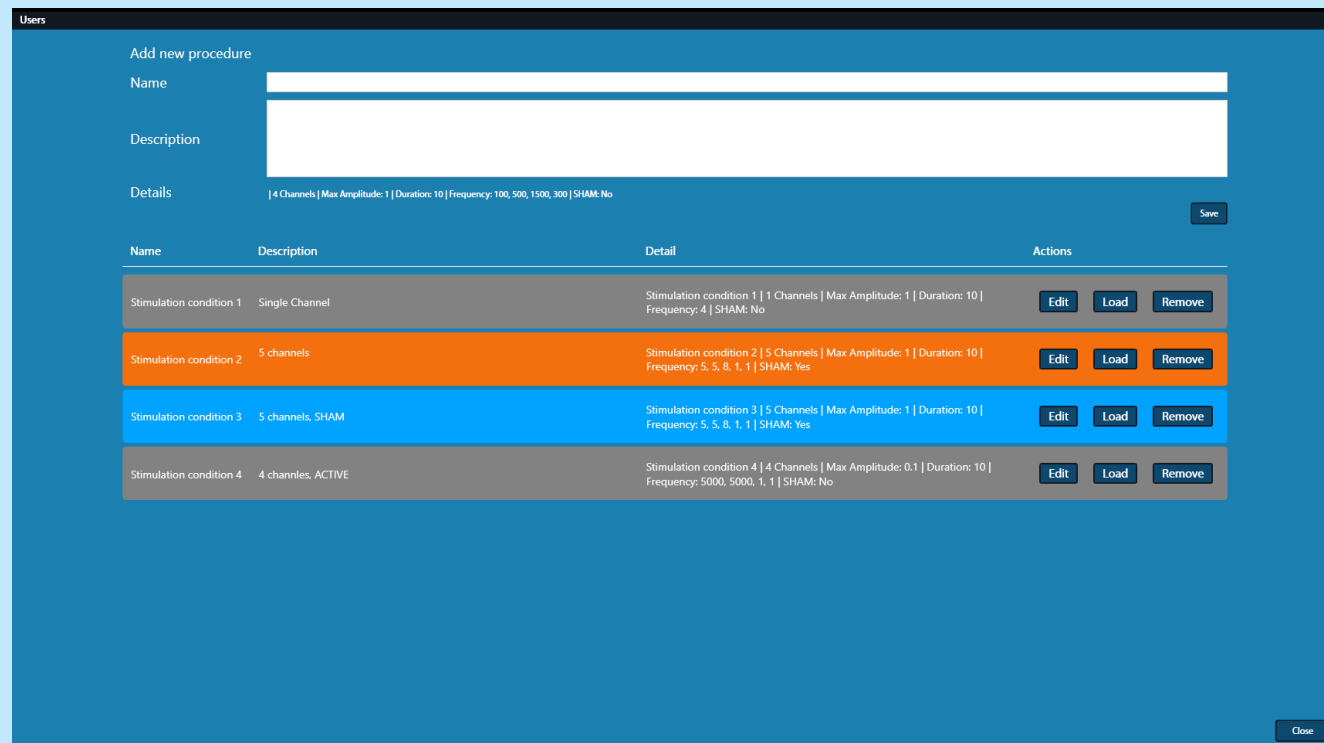
Other visible options include: File, Save Procedure, Logout, Exit.

Clear/Program procedure

Save programed procedure

Load previously saved procedures

Export stimulation details and log



The 'Users' interface displays the 'Add new procedure' form and a table of existing stimulation conditions.

Add new procedure form:

- Name:
- Description:
- Details:
- Save button

Stimulation conditions table:

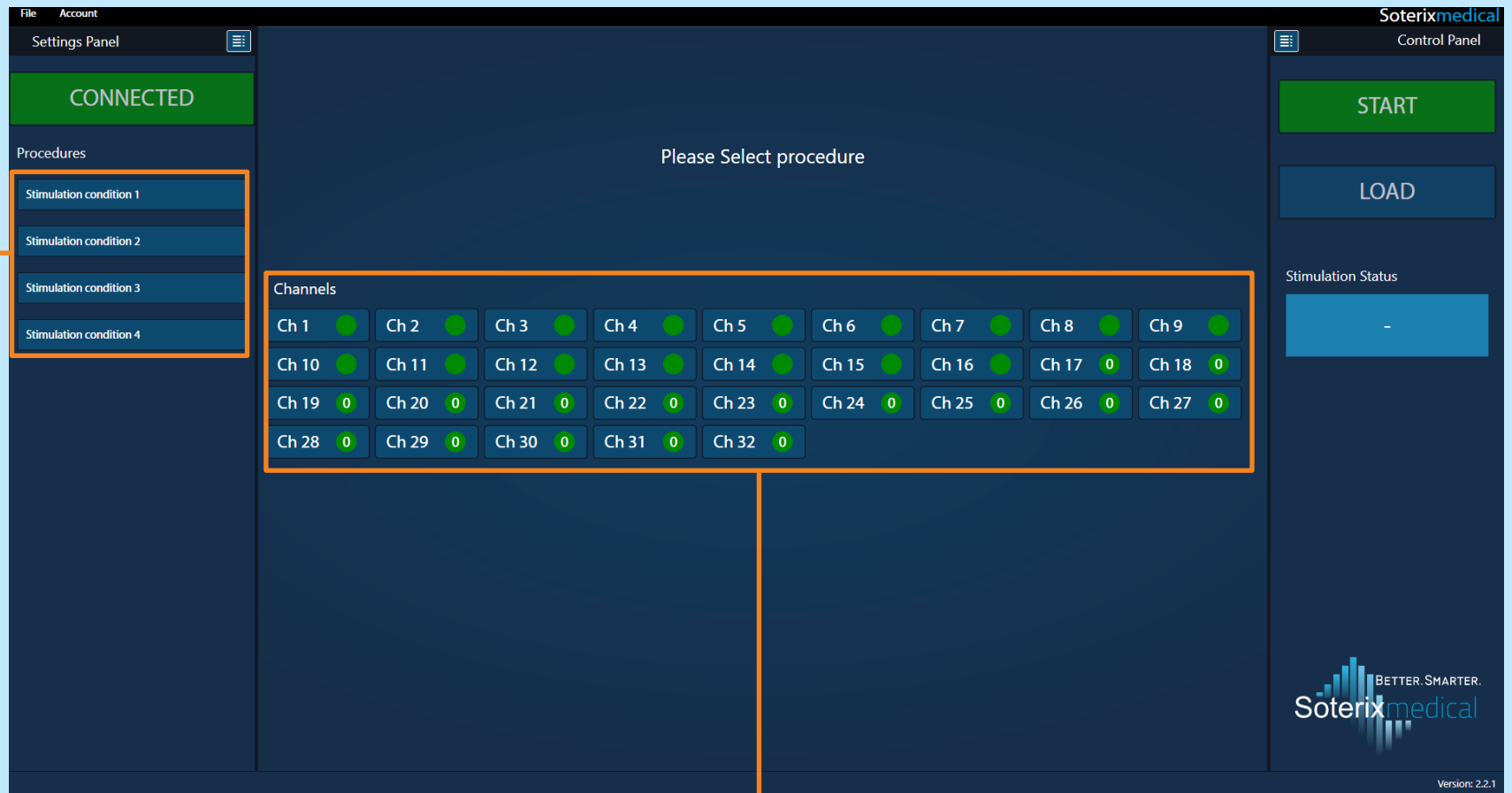
Name	Description	Detail	Actions
Stimulation condition 1	Single Channel	Stimulation condition 1 1 Channels Max Amplitude: 1 Duration: 10 Frequency: 4 SHAM: No	Edit Load Remove
Stimulation condition 2	5 channels	Stimulation condition 2 5 Channels Max Amplitude: 1 Duration: 10 Frequency: 5, 5, 8, 1, 1 SHAM: Yes	Edit Load Remove
Stimulation condition 3	5 channels, SHAM	Stimulation condition 3 5 Channels Max Amplitude: 1 Duration: 10 Frequency: 5, 5, 8, 1, 1 SHAM: Yes	Edit Load Remove
Stimulation condition 4	4 channels, ACTIVE	Stimulation condition 4 4 Channels Max Amplitude: 0.1 Duration: 10 Frequency: 5000, 5000, 1, 1 SHAM: No	Edit Load Remove

Close button

DOUBLE BLIND: OPERATOR

- Default Username: operator
- Password: operator123

Select
procedure



Colored resistance mapping. No
intensity settings are displayed

ADDITIONAL INFORMATION

- Each channel can deliver up to 3 mA
- Over current protection is activated when total injected current exceeds 6 mA
- Device accepts triggers to start and abort stimulation
- To start stimulation via trigger, stimulation settings should be loaded first from the Load button
- Any consecutive odd/even channels (i.e. 1,2 ; 3,4) cannot be set to have different frequencies
- Single channel selection means that the same stimulation settings are applied automatically for all channels
- Multiple channel selection means that individual waveforms can be applied to selected channels
- Additional users can be created under Accounts\Edit Users

Username	Role	Actions
admin	Admin	<button>Reset Password</button> <button>Remove</button>
operator	Operator	<button>Reset Password</button> <button>Remove</button>
		<button>Add User</button>

- Ramp down setting is automatically set to the ramp up value by default, separate values are not possible
- Delay functionality not available. To be included in future releases



Release Notes: soterixmedical.com/HD-SC/release